

NAME: _____ UIN: _____

QUIZ 3: (100 pts) (Close book/closed notes) Show all your work for full credit1.(15 points) State whether the following statements are **True or False (5 pts each)**

_____ A thermodynamic cycle cannot convert heat completely into work.

_____ The change in entropy (ΔS) for a system cannot be less than zero._____ A closed system process that violates the 2nd law of thermodynamics must also violate the 1st law of thermodynamics.

2. (85 points) A stream of superheated steam at 3 bar and 280 °C with a mass flow rate of 400 kg/s (stream 1) is mixed with a stream of liquid water at 3 bar and 20 °C (stream 2) in an adiabatic mixing chamber. The resulting mixture exits as a saturated vapor at 3 bar (stream 3). Assume steady state operation and neglect any changes in kinetic and potential energy. Calculate the mass flow rate of the liquid water (stream 2) entering the mixing chamber, in kg/s.

$$\left| \frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_i \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum_e \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right) \right. \quad \dot{m} = \frac{AV}{v}$$

1 joule= 1 N.m 1 N= 1 kg.m/s²